SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

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ITA0307 – MOBILE COMPUTING

MediaFlow – Adaptive Mobile Multimedia Streaming & Performance Optimization App

Guided By
Dr.S.K . Saravanan
Professor
Simats Engineering

Presented by:
P.Sudharshan Reddy(192372204)
Y.Vamsi Babu (192311145)
Dinesh(192521131)
V. Susidharan 192521387

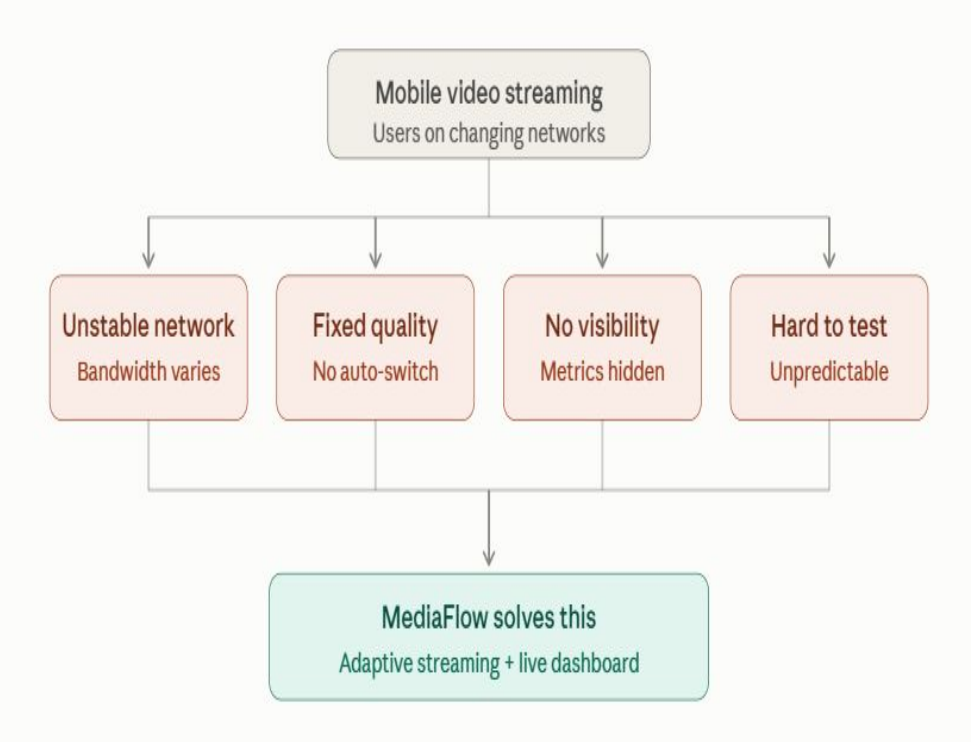
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Abstract

- MediaFlow is a web-based multimedia streaming application designed to provide smooth video playback by adapting video quality according to network conditions.
- The system supports multiple video quality options, allowing users to select resolutions manually or use automatic quality selection based on estimated network bandwidth.
- The application incorporates performance monitoring to track playback status, buffering events, video quality changes, and network performance through an interactive dashboard.
- The proposed system aims to improve the streaming experience, reduce buffering, optimize bandwidth usage, and provide efficient multimedia playback across varying network conditions.

Problem Statement

- **UnstableNetworks:** Changing mobile bandwidth causes buffering and playback interruptions.
- **No adaptive quality:** Fixed-resolution players cannot switch between 360p–1080p or let users choose quality.
- **No performance visibility:** Bandwidth, buffer level, quality switches, and buffering events are hard to monitor.
- **Hard to test:** Real networks are unpredictable, so adaptation logic needs simulated slow, moderate, and fast profiles.



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Existing System

- Fixed Video Quality: Existing basic streaming systems may use a fixed video quality, which can cause buffering when network bandwidth decreases.
- High Bandwidth Consumption: High-resolution video streaming can consume significant internet data, even when lower quality would provide a sufficient viewing experience.
- Frequent Buffering: Unstable network connections can cause playback interruptions and increased video loading time.
- Limited Performance Monitoring: Basic video players may not provide detailed analytics about bandwidth usage, buffering frequency, and streaming performance.

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PROPOSED SYSTEM

- Adaptive Video Quality: Automatically adjusts video quality based on available network bandwidth to provide smoother playback.
- Optimized Bandwidth Usage: Selects a suitable video bitrate to reduce unnecessary data consumption while maintaining good video quality.
- Reduced Buffering: Monitors network conditions and buffer status to minimize playback interruptions and loading delays.
- Real-Time Performance Monitoring: Provides a dashboard to track network speed, video quality, buffering events, and streaming performance using FastAPI and MongoDB.

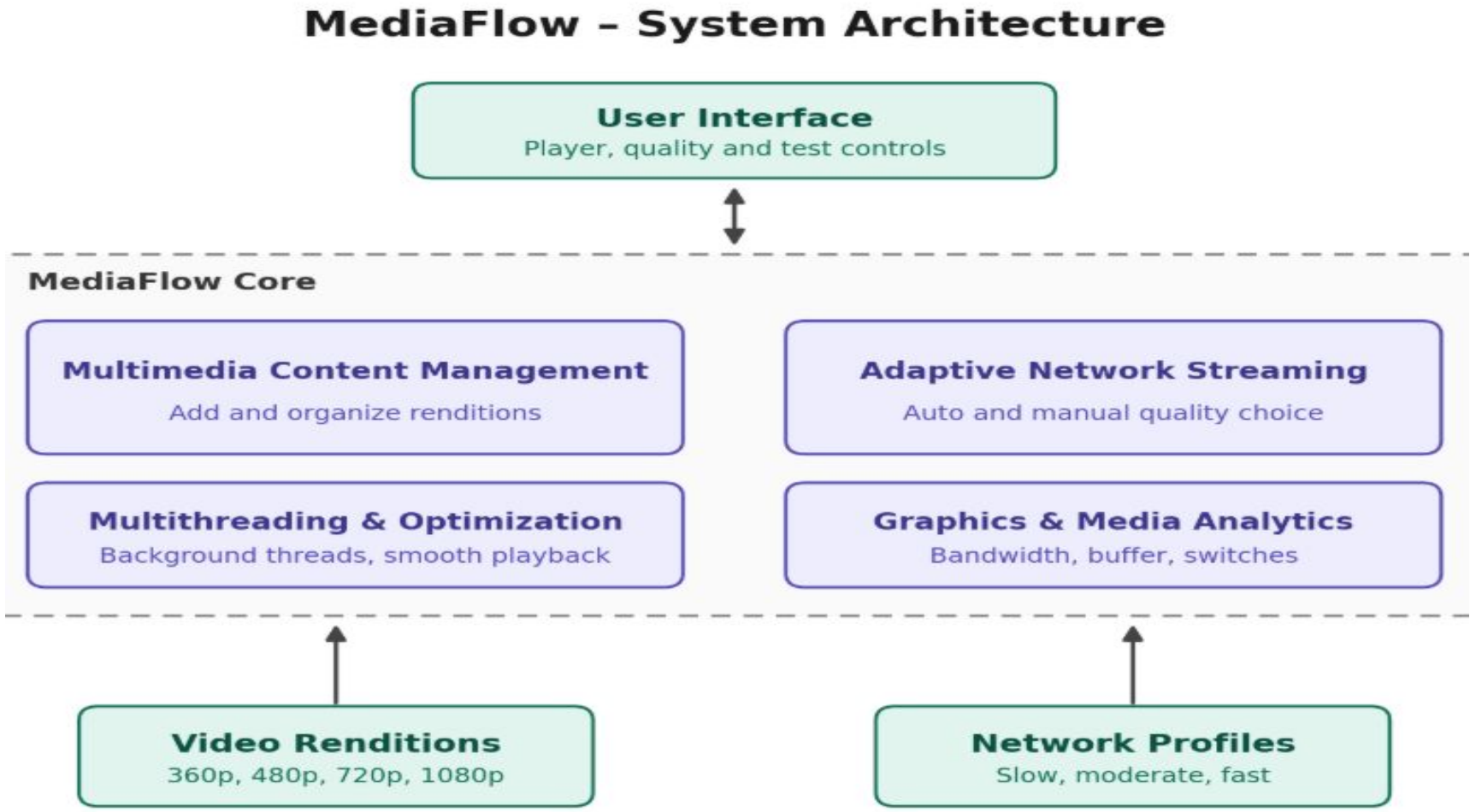
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Literature Review

S.No	Year	Authors	Title of the Paper	Merits	Limitations
1	2025	Christian Timmerer, Hadi Amirpour, Farzad Tashtarian, Samira Afzal, Amr Rizk, Michael Zink, Hermann Hellwagner	HTTP Adaptive Streaming: A Review on Current Advances and Future Challenges (ACM TOMM)	Covers HTTP adaptive streaming from video encoding to end-user consumption, focusing on adaptive bitrate algorithms, quality of experience (QoE), and energy efficiency	A broad review with no implementation. It does not address mobile app design or multithreaded performance tuning
2	2019	Abdelhak Bentaleb, Bayan Taani, Ali C. Begen, Christian Timmerer, Roger Zimmermann	A Survey on Bitrate Adaptation Schemes for Streaming Media over HTTP (IEEE Communications Surveys & Tutorials)	Presents state-of-the-art bitrate adaptation algorithms, which weigh bandwidth estimates, buffer fullness, device features, and viewer preferences	The HAS standards do not mandate any particular adaptation algorithm, so the survey gives no single ready-to-use design. It is also a survey, not a working app
3	2017	Hongzi Mao, Ravi Netravali, Mohammad Alizadeh	Neural Adaptive Video Streaming with Pensieve (ACM SIGCOMM)	Uses reinforcement learning to select bitrates for future chunks, with average QoE improvements of 12%–25% over the best existing schemes	Needs training data and a neural network, which is heavy for a simple mobile app. Later studies have questioned how well it generalizes to unseen networks
4	2017	Yusuf Sani, Andreas U. Mauthe, Christopher J. Edwards	Adaptive Bitrate Selection: A Survey (IEEE Communications Surveys & Tutorials)	Breaks the adaptive bitrate module into resource estimation, chunk request scheduling, and adaptation, giving a clear structure for client-side design	Focuses only on the client-side algorithm. It does not cover multimedia content management, analytics dashboards, or app-level optimization

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System Architecture



Module 1:Multimedia Content Management

- **Multiple renditions:** The user adds separate 360p, 480p, 720p, and 1080p files for the same video. Each file is a version of the content that the player can switch to during playback.
- **Organized library:** Videos are stored and listed in a library, with all the available resolutions grouped under one title, so it is clear which qualities each video supports.
- **Metadata handling:** Details such as video title, duration, file size, and resolution are saved and shown, which helps both the player and the dashboard.
- **On-demand loading:** When the streaming module asks for a quality level, this module provides the matching file quickly, so the switch happens without a long wait.
- **Validation:** The module checks that the files are valid and playable before they are added, which avoids errors during streaming

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Module 2:Adaptive Network Streaming

- **Bandwidth estimation:** The app measures or simulates the current network speed, for example by checking how fast data is loading, to decide what the connection can support.
- **Automatic quality selection:** Based on the estimated speed, the app picks the most suitable resolution. A slow network gets 360p or 480p, and a fast network gets 720p or 1080p.
- **Smooth switching:** The quality changes between renditions without stopping the video, and the playback position is preserved after each switch.
- **Manual control:** The user can ignore the automatic choice and select a resolution, which gives full control when they prefer higher quality or lower data use.
- **Network test profiles:** Slow, moderate, and fast profiles simulate different conditions, so the adaptation logic can be demonstrated and verified without depending on a real network.



Module 3:Multithreading & Performance Optimization



- **Background processing:** Tasks such as buffering, bandwidth checks, and quality decisions run on separate threads, away from the main UI thread.
- **Responsive interface:** Because heavy work happens in the background, buttons, the seek bar, and the dashboard stay smooth while the video plays.
- **Reduced buffering:** Data is loaded ahead of the playback position, which lowers the chance of stalls and long start-up delays.
- **Stable quality:** The logic avoids switching too often when the network speed fluctuates slightly, so the viewer does not see constant quality changes.
- **Efficient resource use:** Memory, CPU, and battery use are kept low, which is important on mobile devices.

Module 4:Graphics & Media Analytics

- **Live performance dashboard:** A panel shows the selected quality, estimated bandwidth, and current playback position while the video plays.
- **Buffer information:** The dashboard displays how much video is buffered ahead, so the user can see how healthy the playback is.
- **Event tracking:** The number of quality switches and buffering events is counted and displayed, which shows how the app reacts to network changes.
- **Real-time updates:** All values refresh continuously during playback, so changes in network profile can be seen immediately.
- **Analysis and demonstration:** The data helps users and developers understand streaming behavior and show that the adaptation logic works as intended.

THANK YOU ….